PharmaLeaf · B.Pharm Semester III · PCI 2026 (NEP 2020)

BP301T Introduction to Machine Learning in Pharmaceutical Sciences

PharmaLeaf notes for B.Pharm Semester III (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order, with concept diagrams, real Python (scikit-learn) code with its actual output, pharmaceutical examples, tables and memory tricks.

Syllabus at a glance

BP301T is a 2-credit core theory course: 2 lectures per week, 30 hours in total, 50 marks (20 sessional + 30 end-semester).

Unit	Title	Topics	Hours
I	Foundations of machine learning	AI, ML and data science; supervised, unsupervised and reinforcement learning; features and labels; training and testing data; model building and prediction; train-test split; overfitting and underfitting; bias-variance trade-off; ML workflow	6
II	Regression models in healthcare	Linear and multiple linear regression; coefficients; residuals and goodness of fit; MSE, RMSE, R^2; sklearn demonstrations (dose-response, dissolution, PK)	6
III	Classification models in clinical applications	Logistic regression; probability and threshold; confusion matrix; accuracy, sensitivity, specificity, precision, recall; ROC and AUC; k-nearest neighbours; ADR and disease-risk examples	6
IV	Tree-based models and ensemble learning	Decision trees (structure, splitting, decision paths, feature importance); random forest and ensembles; advantages and limitations; ADR risk stratification	6
V	Unsupervised learning and case studies	Concept of unsupervised learning; clustering, K-means, choosing k, interpreting clusters; mini case studies: patient segmentation, drug grouping	6

How to use these notes

- Each topic starts with a **definition** (2-mark answers), then an **easy explanation with a pharmacy example**, a **diagram** and a **table** (5- and 10-mark answers).
- Every Python example was **actually run**; the green box below it shows the **real output**. You do not need to memorise code for the theory exam, but you must be able to **read**

and interpret outputs.

- Practice data files (simulated, for learning only) are in the `data/` folder: `pk_clearance.csv`, `dissolution.csv`, `adr_patients.csv`, `patients.csv`, `compounds.csv`.
- The Python basics (variables, lists, Pandas, NumPy) were covered in BP101T and BP201T.

Unit I: Foundations of Machine Learning (6 hours)

I.1 Artificial intelligence, machine learning and data science

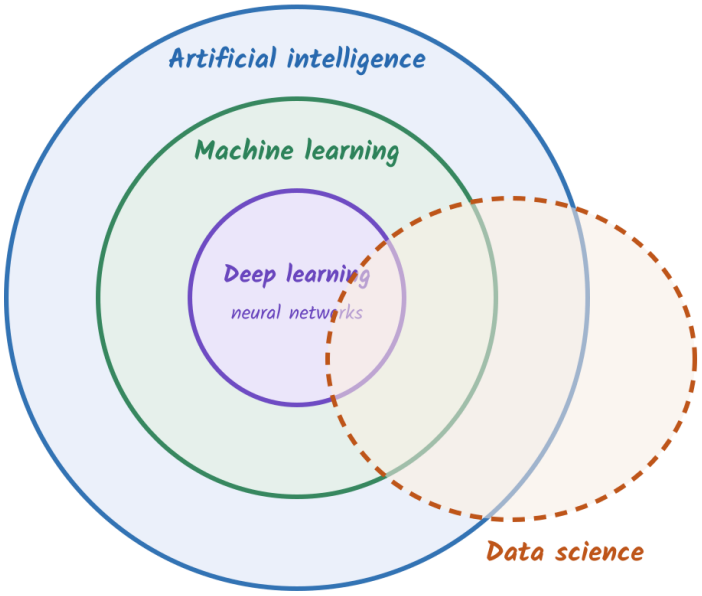
Artificial intelligence (AI): the branch of computer science that makes computers perform tasks that normally need **human intelligence**: learning, reasoning, understanding language, recognising images and making decisions. (Term coined by John McCarthy, 1956.)

Machine learning (ML): a **subset of AI** in which computers **learn patterns from data** and use them to make **predictions or decisions**, **without being explicitly programmed** with rules for every case (Arthur Samuel, 1959). A program learns from experience E for task T if its performance P improves with E (Tom Mitchell).

Deep learning: a subset of ML that uses **artificial neural networks with many layers**; it learns directly from raw data such as images, text and molecular structures (examples: reading X-rays, AlphaFold protein structure prediction).

Data science: an **interdisciplinary field** that uses statistics, programming (Python, R), domain knowledge and ML to **collect, clean, analyse, visualise and interpret data** and extract useful knowledge.

AI, Machine Learning, Deep Learning and Data Science



AI
machines doing tasks that need human intelligence

Machine learning
learns patterns from data, no explicit rules

Deep learning
many-layered neural networks (images, text, molecules)

Data science
statistics + programming + domain knowledge to get insight

Remember: AI > ML > deep learning; data science overlaps all and adds statistics and domain knowledge

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Traditional programming vs machine learning

Point	Traditional programming	Machine learning
Input	**Data + rules** (written by programmer)	**Data + answers** (examples)
Output	Answers	**Rules (a model)** that can give answers for new data
Example	"If CrCl < 30, reduce dose" written as code	Model learns dose-adjustment patterns from thousands of patient records
Handles new patterns	Only if the programmer adds rules	Learns them from new data (retraining)

Scope of AI and ML in pharmaceutical sciences

Area	Applications
Drug discovery	Virtual screening of millions of molecules, predicting activity (QSAR), de novo drug design, predicting ADMET (absorption, distribution, metabolism, excretion, toxicity), protein structure prediction (AlphaFold), drug repurposing